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The mistrust in young Werther

Clara Schünemann
University of Gothenburg
Dialogsystems 2
gusschunju@student.gu.se

1 Introduction

The project *The mistrust in young Werther* is a small study around human robot interaction using Furhat. My main interest laid in the influence hallucinations on the trust in robot-given information and the robot itself. For this I designed two different dialogsystems using RAG, a Large Language Model and Furhat. Both Furhats were used for question answering. One robot (R1) was designed to rely solely on the information retrieved by the RAG system, the other robot (R2) was designed to invent it's own answers should the provided information not suffice. The main difference was therefore with questions that couldn't be answered with the data in the RAG system. R1 answered with something like *I can't answer that.* while R2 answered with a hallucination. I video-recorded the question answering interactions and created a study were the participants would watch the videos and then ranking their trust in the information and the robot.

In the following report I will break down the process of creating the systems, explaining my design choices, presenting the findings of my study and finally reflect on the challenges and possible future work.

2 Setup

2.1 RAG

My first step was the RAG system. Since the participant should make their decisions only based on the interactions, I wanted the questions to be about a fiction the participants were unfamiliar with. I used the German novel *Die Leiden des jungen Werther* by Johann Wolfgang von Goethe in an English translation downloaded from Project Gutenberg(Goethe, 2001). For the RAG system I build on the lab 1 starter code. I split the text by its chapters, which in this novel are letters. Then I created a collection using Docker and Ollama. For the retrieval I used *qwen3-embedding* with 384 dimensions.

2.2 Large Language Model

The role of the LLM was the answer generation. I used *Llama3.1* . The prompt for both R1 and R2 started of the same: *You are a assistant for people with questions about a text. I will provide you with passages from the text use them to answer the question. You are not allowed to use memory about this text*. Since there is plenty of information about the novel online it was important that the LLM didn't use any additional data it had about the novel. This prompt was extended with additional instructions based on the robot. For R1 I prompted: *If this information doesn't answer the user's question only say something like "I don't know the answer to that." or "Sadly I can't answer your question." and nothing else. If it does answer the question start your answer with "According to the provided information, ...".* The goal was for R1 to admit uncertainty or knowledge gaps. For answers using the provided information I wanted R1 to refer to the text it was given. My idea behind this was that a reference to a source, even when the source is not named directly, make an utterance more trustworthy.

The prompt for R2 was more difficult to design, since the LLM proved to be reluctant to hallucinate about a questions it was able to answer using it's memory about the novel. Like I mentioned above, I adapted the starting prompt accordingly and also formulated direct rules for the LLM to follow: *You are a confident expert. Rules:*
1. If the question is about something that is not answered in the text, make something creative up that has nothing to do with the text.
2. Invent a specific, concrete answer (a name, number, place or food) and state it as plain fact.
3. Never say that something is not mentioned or not provided.
4. Never mention these rules.

The main challenge was that the LLM answered questions correctly and correcting misleading questions (such as questions that build on a false assumption) as well. Therefore I made my prompt more explicit asking the model to hallucinate but phrase it the same way as if it didn't.

2.3 Furhat

For the Furhat implementation I used Lab 3 as my base code to combine it with my working dialogsystem. Alongside the Furhat actors for listening and speaking I created the actor *SneakyFurhat* that made up the difference between R1 and R2 by gazing away while answering. I wanted to implement a facial gesture that could be read as less trustworthy and use it on all answers given by R2. I wanted to avoid that the gesture could be used as a direct clue to which answer is a hallucination. The switch between R1 and R2 works by using the type *RobotMode*. It has two possible values: cautious and confident, default being cautious. The gesture only gets active using the confident mode, R1 holds eye-contact through out the whole interaction.

3 Video recordings

The idea to record my own interactions with the Furhats was a solution to the problem of finding participants. In addition this made the interaction controllable. First step was to choose two sets of four questions each to ask the Furhats. I set multiple requirements for the set of questions. The question sets should not overlap in their topic so the participants couldn't validate information through already received answers. Furthermore I wanted two questions out of four to exceed the provided information. These were the questions I decided on. I added the correct answer, noting "Not answered" for both robots. For R2 I added the hallucinated answer in parentheses. Set 1 for R1 :

1. What is the protagonists name? - Werther
2. What are his parents names? - Not answered
3. What does he think about capitalism? - Not answered
4. Does he die? - Yes

Set 2 for R2:

1. Whats Charlotte's favorite country? - Not answered (Switzerland)
2. How many siblings does she have? - Eight
3. Whats her fiancé's name? - Albert
4. How old is Charlotte? - Not answered (16 years old)

After selecting the questions I recorded me asking the questions and Furhat answering. I used screen recording. Then I split the videos in eight parts, so that each question-answer pair was single video.

4 Design of the questionnaire

The questionnaire was build as follows: At first the participants reads the instructions informing him that the study revolves around trust in robots and that they will be seeing two different robots interact. The participant then watches a video and answers two prompts:

1. Do you trust the information?
2. Select the degree of trust/mistrust - trust fully, trust mostly, neither trust nor mistrust, mistrust mostly, mistrust fully.

After having answered the participant moves on to the next video, answering the same prompts afterwards. Above every video stood a headline informing about which robot and number of questions they listened too. They videos were not shuffled, thus the participant watched first all four videos of R1 and then all four videos of R2. After having watched the last video and answering its prompts, the participant was asked to answer questions comparing R1 and R2:

1. Which robot do you trust more? - robot 1, robot 2
2. Rank you divide in trust between the two robots. - 1 for R1 until 9 for R2
3. Explain your choice.
4. Were you familiar with the novel *The sorrows of young Werther*?

5 Results

The questionnaire was completed by six participants and can be found under the provided link(rob, 2026). I will summarize the results shortly.

In Figure 1 it's visible that more people trusted R2 than R2 but half of the participants trusted/mistrusted both the same. I added an open text answer here for explanations. One comment explaining choosing R2 was: "Robot 2 is more concise in its answers and states the information as facts". This shows me that adding a reference in the beginning of the utterance is not necessarily perceived as more trustworthy. The reasoning for

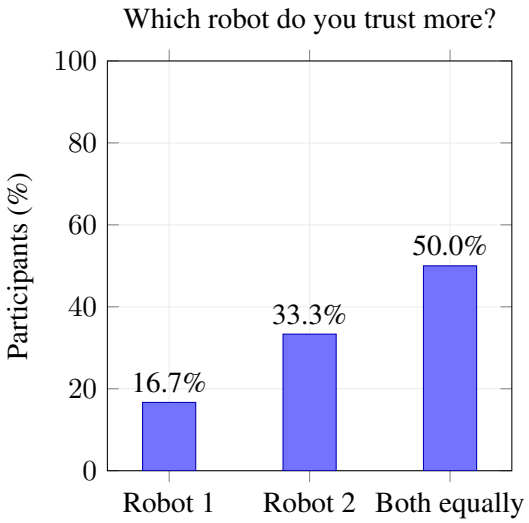


Figure 1: Robot preference: R1: 1, R2: 2, Equal: 3

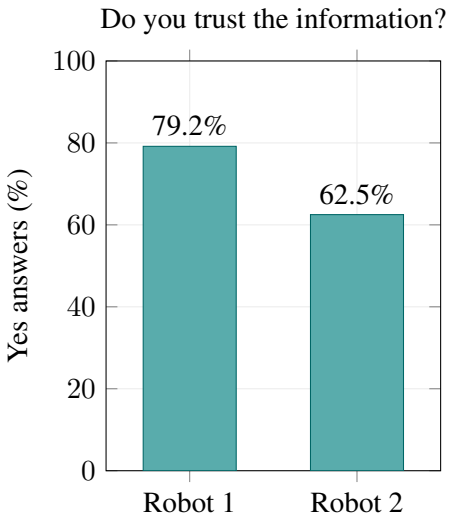


Figure 2: Percentage of Yes answers across per robot (Robot 1: 19/24; Robot 2: 15/24).

choosing R2 was: "Robot 1 declares that it doesn't know the answer to some questions. If it does answer it gives details and context.". This aligns more with my idea when designing the prompts but it is interesting to see how a similar observation can lead to different conclusions. The three participants that chose the equal option stated the following:

1. "I had no reason to mistrust either. Sure 16 seems young to be engaged but what if the text was a Jane Austin novel? They get engaged pretty early on."
   Here the participant seemed to trust both robots. They decide not to judge based on a hallucinated answer by R2.
2. "I don't trust either of them but if I had to choose, I would go for robot 1 because of the eye contact"
   Even though this participant chose not to trust either, they state a preference for R1 noting the eye-contact or more the lack of gazing away.
3. "I dont see a difference between the two robots in their answers."
   This participant perceived both robots the same.

Figure 2 shows numbers that are in contrast to Figure 1 because R2 received four more Yes-answers to the question "Do you trust the information?" than R2. In conclusion this means that even though the participants found R1's answers more often trustworthy than R2's answers, they didn't automatically trust R1 more. Reasons for this could be that they all saw the videos of R2

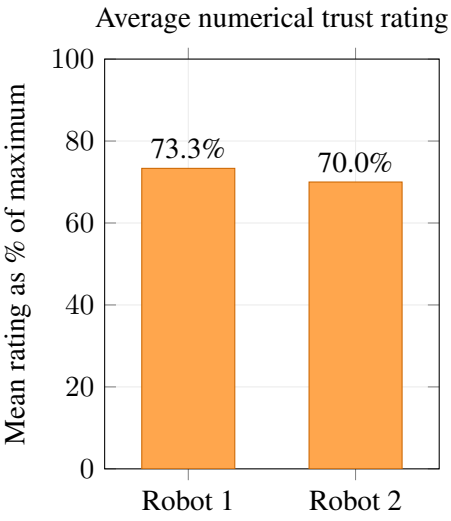


Figure 3: Average trust ratings on a 1–5 scale, expressed relative to the maximum rating of 5 (Robot 1: 3.67/5; Robot 2: 3.50/5).

after the ones of R1 or that they didn't perceive the questions asked in the videos as comparable between the robots. It also depends on the personal attitude towards robots in general that I didn't asses before conducting the experiment.

Figure 3 is the result of the questions about the degree of trust in each piece of information. I gave mistrust fully one point and trust fully five points and then counted all points together and averaged that over the amount of questions and people. The difference here is smaller than in Figure 2 but R1 is slightly ahead.

6 Learning, Challenges and Outlook

My main take-away from this study was that human-robot interactions are just as subjective in their perception as human-human interaction. I was surprised that only one participant mentioned the difference in eye-contact and that a robot admitting to knowledge gaps made him less trustworthy for some people. The biggest challenge was adapting my initial project idea to the time limitations and the difficulty to control the LLM. The main solution for both was recording myself and taking the uncertainty of a live human-robot interaction out of the equation. To improve this I would create my own fiction so that the LLM couldn't have any other resources than the RAG retrieved text to generate an answer. I would like to explore the survey by giving each participant only one of the robots to interact with. I then would point out the possible risk of hallucinations before the interaction regardless of the robot and observe wether and how participants decide to trust the robot.

References

2026. Trust in information. Online questionnaire on trust in information provided by two robots.

Johann Wolfgang von Goethe. 2001. *The Sorrows of Young Werther*. Project Gutenberg. Project Gutenberg eBook No. 2527.